INCH-POUND

A-A-58002A
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION
GENERATOR, FUNCTION-PULSE, ELECTRONIC TEST

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) describes the minimum requirements for a Function Pulse Generator in which only those manufacturers meeting or surpassing the following requirements are supplied per this CID.

2. SALIENT CHARACTERISTICS.

2.1 Classification Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet the following description. The equipment shall be Class 3, in accordance with MIL-PRF-28800, except as specified herein.

2.2 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein except as specified herein.

2.2.1 Temperature Operating. The equipment shall conform to its specified performance and accuracy while being operated in a temperature range of 0°C to +50°C.

Comments, suggestions, or questions on this document should be addressed to WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611. Since contact information can change, you may want to verify the currency of this information using the ASSIST Online database at www.dodssp.daps.mil.

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2.2.2 Temperature Non-operating. The equipment shall conform to its specified performance and accuracy after being stored in an environment in the temperature range of -10°C to +60°C.

2.3 Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 120 VAC (±10 %) at line frequencies of 47 Hz to 63 Hz. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein.

2.4 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 3000 hours of operation with a statistical certainty of 95%.

2.5 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. The calibration adjustments, wherever possible, shall be accessible without removal of the equipment case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2000 hours. No special tools shall be required for alignment or adjustment except those furnished with the equipment.

2.6 Performance Requirements. The equipment shall meet all requirements specified herein, after the warm-up period specified in MIL-PRF-28800, under all combinations of input power conditions, output signal conditions and operating service condition as specified in MIL-PRF-28800.

2.6.1 Frequency.

2.6.1.1 Frequency Range. The frequency range shall be a minimum of 10 mHz to 50 MHz.

2.6.1.2 Accuracy (Continuous Mode). The accuracy of this instrument shall within the following ranges.

10 mHz – 999.9 mHz ± 3%
1.0 Hz - 50 MHz ± 1 %
Jitter: less than 0.1% ± 50 ps

2.6.1.3 Resolution. The frequency resolution of this instrument shall be a minimum of 4 digits.

2.6.2 Output Amplitude.

2.6.2.1 Range. The amplitude range into a 50 ? shall be a minimum of 10 mVpp to 10 Vpp.

2.6.2.2 Resolution. The amplitude resolution shall be a minimum of 3 digits.

2.6.2.3 Amplitude Accuracy. The amplitude accuracy at 1kHz shall be ± 4 % of the setting.

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2.6.2.4 Flatness. The minimum flatness shall be within the following ranges.

10mHz to 999.9 kHz: ± 1%
1MHz to 9.999MHz: ± 2%
10MHz to 50 MHz: - 15%

2.6.3 DC Offset.

2.6.3.1 Range: ± 5 Vpk ac + dc

2.6.3.2 Accuracy (at 1 kHz) shall be ± (1% of setting + 1% of amplitude + 2 mV)

2.6.3.3 Resolution: 3 digits

2.6.4 Harmonic Distortion (Sine). The maximum harmonic distortion shall be as follows.

100mHz to 1MHz: <1% THD
1MHz to 5MHz: Max harmonic < -40 dBc
5MHz to 50 MHz: Max harmonic <-21 dBc

2.6.5 Pulse.

2.6.5.1 Pulse Period. The minimum pulse period shall be as follows.

Range: 20 ns to 99.99 s
Resolution: 4 digits
Accuracy and Jitter: Same as Frequency

2.6.5.2 Pulse Width. The minimum pulse width shall be as follows.

Range:	10 ns to 999 ms
Accuracy:	10ns to 99.9 ns : ± (5% + 2 ns)
	100ns to 999 ms : 3% ± (4% + 2 ns)
Resolution:	3 digits

2.6.6 Ramp.

2.6.6.1 Ramp Period. The minimum ramp period shall be as follows.

Range: 10 µs to 99.99s
Resolution: 4 digits

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2.6.6.2 Ramp Width. The minimum ramp width shall be as follows.

Range: 5µs to 999ms
Accuracy (5µs to 999ms): 3%
Resolution: 3 digits

2.6.6.3 Triangle and Ramp Linearity. Linearity shall be (10% to 90% Amplitude) > 99% at less than 5 MHz.

2.6.7 Amplitude Modulation. The instrument shall have an external amplitude modulation input. The input shall be compatible with the following AM characteristics.

Depth of Modulation Range: 0 to 120%
Modulation Frequency Range: DC to 20 kHz.

2.6.8 Triggering. The instrument shall have an external trigger input. The input shall be compatible with the following characteristics.

2.6.8.1 Impedance: 10 k? ± 5%

2.6.8.2 Maximum Input Voltage: TTL compatible

2.6.8.3 Minimum Pulse Width: 100 ns

2.6.9 Remote Operation. The equipment shall interface with IEEE-488.

2.6.10 Sync Output. The instrument shall have a sync output.

2.6.10.1 Level (at 50 ?). The Sync output level shall be 0 to 1V.

2.6.10.2 Rise/Fall Time. The rise/fall time on the sync output shall be less than < 3 ns.

2.6.11 Output Impedance. The output impedance the equipment shall be compatible with 50 ? .

2.6.3 Physical Characteristics. The test set's component dimensions and weight shall be consistent with current commercial offerings. The equipment shall have the following physical characteristics.

2.6.3.1 Dimensions. The width and depth of the equipment shall be compatible with mounting in an ANSI/EIA-310-D standard equipment rack. The height shall be compatible with three standard height units, 3U. One unit or "U" is defined by ANSI/EIA-310-D to be 1.75 inches or 44.45 mm.

2.6.3.2 Weight. The total weight of the equipment, excluding accessories and manuals, shall not exceed 15 pounds (or 6.8 kilograms). If the depth of the equipment is greater than 24 inches the maximum allowable weight shall be reduced by 33 percent.

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2.6.3.3 Rackmount. Rackmount capability shall be available as an option for the equipment specified herein.

2.6.4 Manuals. The equipment shall be delivered with operator, maintenance, illustrated parts, and calibration manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

4.3 The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Sources of documents.

6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.2 The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.1.3 IEEE standards can be obtained from IEEE Customer Service Center, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331, U.S.A. Phone: 800-678-4333 (Toll-Free, USA & Canada), 732-981-0060 (Worldwide), Fax: 732-981-9667. Website: www.ieee.org.

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6.1.4 ANSI/EIA standards can be obtained from the Electronics Industry Association, Engineering Department, 2001 Pennsylvania Ave., N.W., Washington, D.C., 20006. Phone: 800-854-7179 (USA and Canada).

MILITARY INTERESTS

Custodian:
Air Force – 99
Army – CR
Navy – EC

Preparing activity:
Air Force – 84

Reviewing activities:
DLA - CC

Agent:
Air Force - 99

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